

FRAUD RISK REFERENCE

From Cookie IDs to Agentic AI

Have Fun with Fraud Variables

FIND YOUR SIGNAL — CLICK A WORD TO JUMP TO ITS SECTION

IP Phone Email **Cookie** Device Intelligence

Behavioral Biometrics SSN Address

Payment Instrument Bank Account **Agentic AI**

How IP, phone, email, cookies, device intelligence, behavioral biometrics, SSN, address, payment instruments, and bank accounts power fraud risk intelligence — what to watch for in each variable, who provides the data, and how agentic AI is changing the game.



By **Priyanka Aggarwal** | October 2026

It started with a bust-out group

Twenty-one years ago, I started my fraud journey inside a credit bust-out group — and the fun we had. Pull an IP address or a cookie ID, make clusters out of it, put velocity controls on it, build a negative database out of it: the thrill of the day we realized how much a single signal could tell us is still alive in me.

Over the years the variables kept piling up — phones, emails, devices, behaviors, bank accounts — but the basics never left me: cluster what you can see, watch how fast it's happening, remember what hurt you before.

Now agentic AI is changing the variables fraud leaders are used to seeing. Some signals are quietly going extinct; some are holding their ground. This article is the walk from those hand-built IP and cookie clusters to the agentic era: what each variable tells you, what to watch for in each variable, and the solution providers that turn them into decision-ready intelligence. The vendor matrix near the end puts the landscape on two pages, before a closing section on privacy and fairness. Have fun with fraud variables.

Terms as used here: a **variable** is any fraud-risk intelligence input; an **identifier** is a data element the user presents or the system collects (a phone number, an IP address); a **signal** is derived from identifiers or context (a datacenter IP changed five minutes ago); a **verdict** is what a policy, score, or model returns.

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Privacy and fairness

1. IP address — the first signal you get

The IP address is usually the first piece of intelligence you touch: it tells you where the request entered your systems before the user has typed a single character. Vendors enrich it in milliseconds with geolocation, connection type, reputation, and velocity context.

What it tells you

- **Geolocation** — country, city, and usually a radius.
- **Connection type** — residential ISP, mobile carrier, datacenter, hosting provider; datacenter traffic on consumer flows has historically been suspicious, though agent-driven shopping traffic legitimately runs on cloud ASNs now (see §12).
- **Anonymity detection** — VPN, proxy (commercial, residential, or rotating), and Tor exit nodes.
- **Reputation and linkage** — whether the IP has appeared in fraud before, and how many distinct accounts, cards, or identities have used it.

Geolocation consistency

IP geolocation is only half the story; cross the IP's location against billing, shipping, and phone geography plus device timezone. Matches raise confidence; each conflicting hop should cost trust.

- **The four-corner check** — IP geo versus billing geo versus shipping geo versus phone geo; no single one is authoritative, but together they catch the laziest lies.
- **Impossible travel** — two logins or transactions too far apart in space and time to be humanly possible.

Things to watch for

- **Velocity** — many accounts, card tests, or applications funneling through one IP — classic account-creation and card-testing farms.
- **Anonymity stacking** — datacenter IP plus Tor exit plus inconsistent geo is a different animal than a single-family VPN.
- **The VPN caveat** — many legitimate users run VPNs; treat anonymity as a risk weight, not a verdict — and never blocklist an IP forever: IPs cycle fast.

Solution providers

- **IPQualityScore (IPQS)** — IP reputation, proxy/VPN/Tor detection, plus device and phone signals.
- **MaxMind** — the geolocation and risk-score reference many vendors build on.
- **LexisNexis ThreatMetrix** and **Fingerprint** — IP context inside broader device-identity verdicts (Fingerprint's "Smart Signals" include VPN/proxy detection).

2. Phone number — identity anchor and authentication channel

A phone number plays two roles at once: it is the channel through which one-time passcodes arrive, and it is an identity signal with tenure, ownership, and risk history. Fraud that defeats the second puts the first at risk.

What it tells you

- **Validity and type** — valid format, line type (mobile, landline, VoIP, toll-free), carrier, and geography.
- **Network state (HLR lookup)** — live carrier data: is the phone on, roaming, or ported?
- **Tenure** — how long the number has existed and been associated with this person; a six-day-old number on a new credit application is a neon flag.
- **SIM swap / port recency** — was the SIM changed or the number ported recently? The single highest-value phone signal: it directly precedes OTP interception.
- **Linkage** — how many accounts, identities, or cards are tied to one number.

Things to watch for

- **VoIP and disposable numbers** — cheap, anonymous, replaceable; skewed heavily toward fraud even though legitimate use exists.
- **Fresh numbers** — new numbers tied to brand-new accounts are the phone equivalent of a fresh device.
- **SIM swap within hours or days** — swap, then password reset, then funds movement is the account-takeover sequence.
- **Number changed on the account** — fraudsters swap the phone number first so the OTPs come to them.
- **OTP delivery failures** — a number that cannot receive codes is either dead or synthetic.

Solution providers

- **Telesign** — phone verification, risk scoring, SIM-swap detection.
- **Prove** — phone-centric identity: tenure, SIM swap, phone-to-person matching.
- **Ekata (a Mastercard company)** — phone-to-identity linkage in transaction risk.
- **Socure** — phone risk signals within its identity graph.
- **iconectiv** — the source of truth on US number portability.
- **TransUnion (via Neustar), Twilio Lookup** — carrier, line type, and lookup at scale.

3. Email — the identity signal you already collect

Email is the oldest digital identity claim on file and usually arrives for free with the transaction — and its fraud story has quietly become one of the richest. Vendors answer three questions: is this address real, deliverable, and old; does it cohere with the person claiming it; and how much history it carries in the wider ecosystem.

What it tells you

- **Tenure** — the first-seen date of the address; an inbox born minutes ago on a lending application behaves like a fresh phone number or a new device. Age alone is not proof — aged email accounts are bought and sold on underground markets.
- **Domain class** — disposable or temporary-mail domains, free personal-mail providers, corporate domains, newly registered or parked domains, and typosquat lookalikes of trusted names.
- **Deliverability** — the mailbox exists and accepts mail; fabricated identities frequently die at confirmation time.
- **Coherence** — the local part should rhyme with the claimed name (janedoe@ versus xqz7731@) — total incoherence invites a second look.
- **Breach exposure** — the address appears in leaked credential sets, which raises account-takeover odds on any password-resettable flow.
- **Linkage** — how many applications, cards, or payout destinations share one inbox.

Things to watch for

- **Disposable mail on high-trust actions** — convenience for testers, reusable infrastructure for fraud rings spinning up armies of accounts.
- **Subaddressing at scale** — user+01, user+02 harvesting referral bonuses or free trials from a single inbox; the volume is the flag, not the technique.
- **Email changed on the account** — a new inbox swapped in ahead of a payout follows the same playbook as the SIM-swap account-takeover sequence.
- **Mailbox that cannot receive** — no confirmed contact possible; synthetic identities often arrive exactly this way.

Solution providers

- **AtData (now part of Experian)** — the email-centric identity and fraud platform, built from the TowerData and FreshAddress merger, acquired by Experian in February 2026.
- **LexisNexis Risk Solutions (Emailage)** — email-centric risk scoring from a global email network.
- **SEON** — email lookup modules alongside IP, phone, and social-signal lookups.
- **Socure, Equifax, TransUnion** — email signals inside the broader identity graphs.

4. Cookies — the persistence anchor

Cookies are the cheapest persistence the fraud stack has: a first-party cookie dropped during a clean session lets you recognize the same browser when it comes back. It does not authenticate anyone — it only answers "have we seen this browser before, and was it clean?"

What it tells you

- **Session continuity** — a cookie issued on a trusted session turns a returning visitor into a known quantity.
- **Linkage at low cost** — multiple accounts sharing one cookie ID is account-farming arithmetic.
- **Cookie age versus account age** — misalignment cuts both ways: a years-old cookie on a one-day-old account is suspicious, and a brand-new cookie on a years-old account is the classic takeover shape.

Things to watch for

- **No cookies at all** — automation tooling and headless browsers frequently arrive without any.
- **Cookie churn** — an account that sheds its identifiers every session is either privacy-conscious or hiding something — pair it with other signals before deciding.
- **Third-party cookie decline** — cross-site identifiers are fading from the web; never build a control on them again.
- **Cookie theft** — session cookies are a takeover vector; aggressive cookie-based trust without binding to device and IP backfires during cookie-hijacking attacks.

The durable pattern: treat cookies as one edge in the linkage graph, not a device ID. Serious device recognition survives cookie clears and incognito sessions — that is the next section's job.

Solution providers

- **First-party (your own cookies)** — persistence is usually built in-house, complemented by the device-intelligence providers in the next section.
- **Sift, IPQualityScore, TransUnion, LexisNexis** — browser persistence and account-association scoring inside broader risk platforms.

5. Device intelligence — who is really holding the device

Device intelligence answers a different question than cookies do: not "have we seen this browser session," but "what is this piece of hardware, is it trustworthy, and what has it done elsewhere?" It splits into three layers: attributes, integrity, and reputation.

What it tells you

- **Hardware and software attributes** — on mobile: device model, OS, sensor inventory, secure-hardware attestation; on browsers: canvas and WebGL fingerprints, screen, language, and timezone.
- **Integrity** — is the device rooted or jailbroken, is it an emulator, is the app the genuine one or a clone, has location been spoofed? Every software-visible identifier is writable on a rooted device — hardware attestation verdicts (Play Integrity, App Attest) remain the tell.
- **Reputation** — has this device identifier been seen in fraud at other organizations in a shared data network?

Things to watch for

- **New device, new money** — new device plus first transaction plus high-risk action is the composite that matters.
- **Emulators** — hollow sensor inventories and spoofable build properties; legitimate for developers, toxic at checkout.
- **Attribute inconsistency** — a "brand-new" device that reports three different RAM sizes across vendors is showing its seams.
- **Device farms** — dozens of near-identical device signatures renewing accounts or harvesting promos.

Solution providers

- **LexisNexis ThreatMetrix** — the canonical device-identity network with cross-organization device reputation.
- **Fingerprint** — developer-grade persistent device ID (survives cookie clears and incognito) with named "Smart Signals."
- **SHIELD** — device-first platform with emulator, cloned-app, and GPS-spoof detection.
- **TransUnion (iovation)** — device reputation across fraud consortium data.
- **Incognia** — mobile location intelligence fused with device signals.

6. Behavioral biometrics — how they interact, not just what they claim

If device intelligence watches the hardware, behavioral biometrics watches the human: typing cadence, keystroke dwell time, mouse movement, swipe pressure, gyroscope handling — thousands of micro-signals collected passively during a session. The model builds a profile of "how this user's body works" and scores each session against it.

What it tells you

- **Continuous authentication** — the session's interaction signature matches the enrolled user, or it does not.
- **Malware and remote-access detection** — sessions driven through remote-access tools carry giveaway signatures (scripted mouse paths, impossible consistency).
- **Bot versus human versus hybrid** — vendors now explicitly classify whether a session is led by a human, a human-plus-agent hybrid, or an autonomous agent.
- **Scam and coercion markers** — hesitation, fumbling, mid-session pauses, and fields typed by two "people" are the signatures of a victim being coached in real time.

Things to watch for

- **Behavior drift at high-value moments** — the same user behaves differently when wiring their life savings — to a scammer, or under the scammer's remote control.
- **Paste-heavy sessions** — credentials pasted instead of typed, especially with a foreign device.
- **Environment mismatches** — login-page behavior that does not match the enrolled profile after a credentials-only login is an account-takeover call.

Market note: this sector is consolidating — Visa acquired the category pioneer **BioCatch** in 2026, nearly a decade after Mastercard acquired **NuData**. Both card networks now treat behavioral intelligence as core payments infrastructure.

Solution providers

- **BioCatch (Visa)** — the category pioneer; continuous monitoring for 350+ banking clients.
- **NuData (Mastercard)** — behavioral and device intelligence for account takeover and registration fraud.
- **BehavioSec (LexisNexis)** — keystroke and mouse dynamics inside the ThreatMetrix ecosystem.
- **TypingDNA, Experian NeuroID, ThreatMark** — focused behavioral-signal specialists.

7. SSN — the anchor behind synthetic identities

In US fraud work the Social Security number is the anchor identity claim — but its real fraud fight is no longer simple forgery. The question vendors answer now is: *does this name, date of birth, and SSN cohere as one real person, or was this identity assembled?*

What it tells you

- **Deterministic match** — the Social Security Administration's electronic Consent Based SSN Verification (eCBSV) confirms whether a submitted name, DOB, and SSN match SSA records — the definitive check.
- **Internal consistency** — SSN issuance patterns and ranges versus the claimed date of birth; a number issued *before* the claimed birth, or suspiciously recently for an adult applicant, flags an assembled identity.
- **Death-list cross-checks** — SSNs of the deceased reused in applications.
- **Velocity and reuse** — the same SSN behind many applications, or a number that suddenly reappears after years of silence.

Things to watch for

- **Synthetic identities** — real SSN plus fabricated name, or vice versa; the fastest-growing identity-fraud vector in lending.
- **SSN randomization (post-2011)** — SSA began randomizing SSNs in 2011: area-number geography checks are obsolete, and issued-date checks work reliably only for pre-2011 numbers — use eCBSV and mismatch logic.
- **Thin-file fairness trap** — legitimate young consumers and new arrivals also look thin; the best scores separate "new to credit" from "fabricated."
- **Fragmentation risk** — one real person whose applications carry typos or aliases can split into "synthetics" inside your own data.

Solution providers

- **SSA eCBSV**, accessed through permitted entities and resellers — the definitive name-DOB-SSN match.
- **SentiLink** — Synthetic Fraud Score plus eCBSV access; purpose-built to catch identity incoherence at application time.
- **Socure** — Sigma Synthetic Fraud and a full identity platform; widely used by large US banks.
- **Equifax, Experian, TransUnion** — SSN matching and identity verification against bureau files.

8. Address — verification plus velocity

The address earns its place three independent ways: as a verification input against the issuer's records, as a linkage node connecting applications, and — least exploited — as a delivery-velocity signal.

What it tells you

- **AVS match** — the Address Verification System returns a result code from the issuer comparing the billing-address numbers to the card's file; a full mismatch on a card-not-present transaction is a decline instinct.
- **Hygiene and validity** — is the address real (USPS-validated), deliverable, residential or commercial? Lying in street format is its own signal. Standardize addresses before matching — minor formatting differences will create false links and missed ones.
- **Identity linkage** — the same address across many applications, phone numbers, or names — shared addresses behind ostensibly unrelated identities.
- **Delivery velocity** — how many packages are heading to one address across senders; addresses absorbing order spikes are staging grounds.

Things to watch for

- **AVS partials** — ZIP matches but street fails, or vice versa; partial matches price differently by merchant category.
- **Freight forwarders and reshipping mules** — orders that clear every check and ship to a forwarding address.
- **Address change plus order spike** — new shipping address added, followed immediately by high-value orders.
- **Drop-address concentration** — many orders to a vacant, commercial-but-residential, or short-term rental address.
- **Passing validation ≠ real resident** — an address can clear CASS deliverability checks and still be fabricated; deliverable is not the same as lived-in.

Solution providers

- **Issuer AVS** (card networks) — the authorization-time billing-address check.
- **Smarty, Melissa** — USPS-grade address validation, hygiene, and enrichment.
- **LexisNexis, Experian, Equifax, Socure** — address-to-identity verification against reference data.

9. Payment instrument — BIN, card, and wallet intelligence

The card or token carries its own intelligence layer: issuer, country, product type, and attack history, with authentication results arriving in the authorization response.

What it tells you

- **BIN/IIN profile** — network, issuing country, issuer, and product type: credit, debit, or prepaid — with prepaid and non-reloadable cards historically over-indexing for fraud.
- **Card tenure** — how recently the card was issued and how long it has behaved normally in-network; a day-old card buying a laptop is a first-purchase story.
- **Presentation context** — raw PAN versus a wallet-network token (Apple Pay, Google Pay): token enrollment is itself an authentication event.
- **Authentication outcomes** — AVS and CVV results, plus 3-D Secure 2.x outcomes where enrolled; score the CVV *match*, not the value, since CVV data can never be stored; a passed 3DS challenge shifts liability to the issuer.
- **Attack footprints** — enumeration bursts; one card across many accounts, or one account cycling many cards (I have seen up to 100 on a single account), is the mule shape.

Things to watch for

- **Prepaid-heavy funding on new accounts** — gift cards and reloadable prepaid used to distance the identity from the money.
- **Issuing-country mismatch** — card issued in one country, billing set in a second, shopping from a third; price it as a weight, not a verdict.
- **Fresh card, large first purchase** — sometimes legitimate, so stack it with the other signals before deciding.
- **Wallet switch** — the same card added to a fresh wallet on a new device hours before the purchase is an account-takeover shape.

Solution providers

- **Visa** — VAAI Score for enumeration attacks; **Ethoca (Mastercard)** and **Verifi (Visa)** for pre-dispute alerts; BIN/IIN APIs for issuer, country, and product.
- **Riskified, Forter, Signifyd** — e-commerce decision platforms, several with chargeback-guarantee models.
- **Accertify, Kount (Equifax), Sift** — consortium-backed transaction risk scoring.

10. Bank account — verifying the rails behind ACH and payouts

Every ACH debit, account-to-account push, and marketplace payout touches a bank account — and the question is always ownership: does this account belong to the person claiming it?

What it tells you

- **Account-ownership validation** — do the name, routing number, and account number cohere as one real holder — the root question behind R10 (not authorized) returns and Nacha's 0.5% unauthorized-return ceiling. Nacha rules require validating bank accounts the first time they are used for online debits.
- **Account age** — newly opened accounts funding payouts or large debits are the bank equivalent of fresh devices.
- **Return history** — R01 (insufficient funds) and R10 (not authorized) signatures per account; originator-level return rates are a compliance clock as much as a fraud meter.

Things to watch for

- **Payout account added, then drained fast** — the defining marketplace and lending payout scam: add, verify lightly (even micro-deposits get passed), pull, disappear. Watch the first days of any added account's life.
- **Account-title mismatch** — the account's title differs from the customer of record; route edge cases (joint accounts) through real ownership validation.
- **R10 spikes on your own debits** — trace the unauthorized-return attack back to how the payer was validated.
- **Mule accounts** — real accounts recruited or rented: inbound credits followed by fast outbound transfers.

Solution providers

- **GIACT (LSEG)** — account-ownership validation and ACH verification (gVERIFY / gAUTHENTICATE), used by processors and originators.
- **Early Warning Services** — bank-consortium deposit-account and payment-risk data.
- **Plaid (Signal)** — ML risk engine scoring ACH return and fraud risk on bank connections.
- **Finicity (Mastercard)** — open-banking account verification and ownership data.

11. Making the variables work together

Ten variables, but the winning move is not any one of them — it is the graph they form. A new device, a fresh VoIP number, an SSN inconsistent with the date of birth, and a first transaction to a drop address are each merely suspicious; together, they are a conclusion. Fraud teams win by correlating, not by collecting.

The linkage principle

Every event should join to history on each identifier: how many accounts has this phone touched? This device? This SSN? This address? The counts, plus their recency, are the cheapest and most decisive features in most fraud models — and the simplest forensic story to tell a reviewer, a regulator, or a jury.

The orchestration layer

Few fraud teams wire ten vendors' APIs directly into a checkout flow. Risk-orchestration platforms sit above the signal providers, routing each event to the right vendors, enforcing the policy ladder (approve, step up, review, decline), and logging the why. Representative platforms: **Alloy**, **FICO**, **Feedzai**, and **Featurespace (Visa)**.

The cost discipline: cheap signals (geolocation, line type, cookie checks) run on every event; passively collected signals such as device reputation and behavioral biometrics layer on continuously; the expensive, high-friction ones (identity verification, SIM-swap queries) fire only at step-up moments. The architecture follows the cost curve.

Governance before you deploy

Device and behavioral signals carry privacy obligations: biometric processing is regulated under laws such as Illinois BIPA and the EU GDPR, and consent, retention, and purpose-limitation rules differ by market. Two principles keep programs honest: use the variable because the fraud decision requires it, not because the vendor sells it, and keep measuring false positives by customer segment — thin-file and new-to-market consumers pay first for sloppy thresholds.

12. How agentic AI is changing these variables

Every variable above was designed for one actor per session: a human at a device. Agentic commerce adds a second actor — an AI agent transacting as the human's *delegate*. Some signals go flat, some matter more, and a new identity layer is being built to fill the gap.

Signals that weaken

IP address. An agent runs in a provider's cloud, so geolocation and ASN context describe a datacenter, not a customer; a shopping agent on a cloud ASN is the expected pattern. The defense moves from IP reputation to signed request identity verified at the edge.

Device intelligence. There is no human hand on a phone; the "device" is an ephemeral container. Root/jailbreak checks go dark, replaced by credentialed agent identity (see below).

Cookies. Agents don't browse like returning customers — sessions may be shared across users or discarded each task. Cookie persistence was always weak; agentic traffic settles the argument for good.

12. How agentic AI is changing these variables — continued

Signals that strengthen

Phone. Step-up authentication must reach the *human*, not the agent session; SIM-swap recency stays the highest-value phone signal.

Address. When the shopper is software, where the goods ship is one of the few anchors left — drop-address linkage, first-order-to-new-address, and delivery velocity matter *more*, not less.

SSN and identity proofing. Agents can assemble synthetic-identity applications programmatically, at machine scale. Enrollment-time verification and synthetic-identity scoring take on more of the total load.

Behavioral biometrics gets a successor: agents don't type or click, so classic scores give way to *agent-behavior profiling* — tool sequences, retry cadence, cart mutation velocity.

Signals that hold steady

Email. Enrollment, verification, and step-up messages still have to reach the human behind the agent; an inbox changed ahead of disbursement stays an account-takeover tell. As agents send mail on owners' behalf, sender authentication (DMARC/SPF) and domain-class checks gain weight as fraud signals.

Payment instrument. BIN profile, prepaid flags, and issuing-country mismatches survive intact — they are card-level, not device-level. The wallet network token the agent transacts with becomes the trust anchor: issuance recency, spend caps, and merchant allowlists turn token metadata into first-class fraud features.

Bank account. Ownership validation and return history change little by channel — a bank account is verified, not browsed. Agent-initiated payouts raise the stakes on payee and payout-destination changes: step up the change itself, not just the transaction.

12. How agentic AI is changing these variables — the new assessment model

Assessment shifts from device trust to delegation trust

Signed, credentialed agent identity is already live: **Visa's Trusted Agent Protocol** and **Mastercard Agent Pay** (with **Verifiable Intent** describing the user's grant) carry live transactions alongside **Google AP2** mandates and **Stripe's Agentic Commerce Suite**. Risk decisions split into two verdicts: *is this agent legitimate* and *does this action match the grant*. The open question for 2026–27 is liability: dispute rules for agent-delegated transactions are still being written.

AI on the defense side

The same technology helps fraud teams: models that link accounts across huge graphs, assistants that summarize cases and draft investigations, and faster adaptation when attackers change tactics. The teams that win will treat agentic AI not as a threat to block outright, but as a new channel to verify — with good agents welcomed and bad ones made expensive.

Vendor landscape at a glance

Representative solution providers by variable — most vendors span multiple columns.

Variable	Representative providers	Decision role
IP address	IPQualityScore, MaxMind, Fingerprint, LexisNexis ThreatMetrix	Geo, anonymity, reputation, velocity gating on every event
Phone number	Telesign, Prove, Ekata (Mastercard), Socure, iconectiv, TransUnion, Twilio Lookup	Type/tenure always; SIM-swap and risk scoring at step-up
Email	AtData (Experian), LexisNexis Emailage, SEON, Socure, Equifax, TransUnion	Tenure, domain class, deliverability; linkage at onboarding
Cookies	Sift, IPQS, TransUnion, LexisNexis (first-party persistence)	Return-recognition and linkage; one edge, not a device ID
Device intelligence	LexisNexis ThreatMetrix, Fingerprint, SHIELD, TransUnion iovation, Incognia	Device risk verdicts and cross-industry reputation
Behavioral biometrics	BioCatch (Visa), NuData (Mastercard), BehavioSec (LexisNexis), TypingDNA, NeuroID, ThreatMark	Continuous in-session risk, ATO and scam detection

Vendor landscape at a glance — continued

Variable	Representative providers	Decision role
SSN	SSA eCBSV, SentiLink, Socure, Equifax, Experian, TransUnion	Deterministic match plus synthetic-identity scoring at onboarding
Address	Issuer AVS, Smarty, Melissa, LexisNexis, Socure, bureaus	Billing match at auth; linkage and delivery velocity
Payment instrument	Visa (VAAI), Ethoca (Mastercard), Verifi (Visa), Riskified, Forter, Signifyd, Accertify, Kount, Sift	BIN profile, prepaid flags, 3DS outcomes, wallet-token context at auth
Bank account	GIACT (LSEG), Early Warning, Plaid (Signal), Finicity (Mastercard)	Ownership validation, account age, return and NSF history at funding and payout
Orchestration	Alloy, FICO, Feedzai, Featurespace (Visa)	Routes events to vendors; approve / step-up / review / decline

Note: provider names are illustrative examples from the market, not endorsements — capabilities and ownership change, so verify current offerings before relying on them. Vendor landscapes shift quickly; evaluate each provider against the decision you need it to make, demand evidence of fraud-loss reduction, and re-test the numbers annually.

Privacy and fairness

Collecting and using these signals comes with responsibility. Be clear with customers about what you collect and why, follow GDPR, CCPA and sector rules such as the FCRA, keep data only as long as needed, and test models for unfair impact on protected groups. Every adverse decision should be explainable. Good fraud programs protect customers twice: from criminals, and from being wrongly treated as one.